

2 Description

The STM32C562xx devices are general purpose microcontrollers family (STM32C5 Series) based on the high-performance Arm® Cortex®-M33 32-bit RISC core. They operate at a frequency of up to 144 MHz.

The Cortex®-M33 core features a single-precision floating-point unit (FPU), that supports all the Arm® single-precision data-processing instructions and all the data types.

The Cortex®-M33 core also implements a full set of digital signal processing (DSP) instructions and a memory protection unit (MPU) that enhances the application security.

The devices embed high-speed memories (512-Kbyte flash memory and 128-Kbyte SRAM), and an extensive range of enhanced I/Os, peripherals connected to three APB buses, three AHB buses, and a 32-bit multi-AHB bus matrix.

The devices feature several protection mechanisms for embedded flash memory and SRAM: readout protection, write protection, and hide protection areas.

The devices embed several peripherals reinforcing security:

- HASH hardware accelerator
- True random number generator
- AES coprocessor

The devices offer two 12-bit ADCs, one DAC channel, one comparator, a low-power RTC, two 32-bit general-purpose timers, two 16-bit PWM timers dedicated to motor control, four 16-bit general-purpose timers, two 16-bit basic timers, and one 16-bit low-power timer.

The devices also feature standard and advanced communication interfaces such as:

- Two I²Cs
- One I³C shared with I²C
- Three SPIs with muxed full-duplex I2S
- Three USARTs, two UARTs, and one low-power UART
- One FDCAN
- One USB full-speed

The devices operate in the –40 to +125 °C (+140 °C junction) temperature ranges from a 2.7 to 3.6 V power supply.

A comprehensive set of power-saving modes allows the design of low-power applications.

The devices offer multiple packages from 32 to 100 pins.

See [Table 1](#) for the list of peripherals available for each part number.

Table 1. Device features and peripheral counts

Peripherals		STM32C562KET	STM32C562KEU	STM32C562CET	STM32C562CEU	STM32C562RET	STM32C562RETxJ	STM32C562MET	STM32C562VET
Flash memory (Kbytes)		512							
SRAM (Kbytes)		128 (including 64 with ECC)							
ICACHE (Kbytes)		8							
Flash memory area for EEPROM Emulation (Kbytes)		64							
One-time-programmable (Kbytes)		4.5							
Timers	General purpose	6							
	Advanced-control	2							
	Basic	2							